

1. Computer System Organization

Think of **System Organization** as the physical layout of an office. It describes how the hardware components are connected and talk to each other.

- **CPU (The Brain/Boss):** Does all the thinking and processing.
- **Memory / RAM (The Desk):** Where the CPU keeps the files it's working on *right now*. It's fast but temporary.
- **Storage / Hard Drive (The Filing Cabinet):** Where data is kept permanently. It's slower but holds a lot.
- **I/O Devices (Inbox/Outbox):** Keyboards, mice, and monitors. How you talk to the computer and how it talks back.
- **System Bus (The Hallway):** The wires that connect everything together so data can travel back and forth.

2. Computer System Architecture

If organization is the office layout, **Architecture** is how many "brains" or workers the office has to get the job done.

- **Single-Processor System:** One CPU doing everything. Like a one-person business. (Standard in older, basic PCs).
- **Multiprocessor System:** Multiple CPUs sharing the same memory and clock. Like a team working at the same large desk. It's faster and more reliable (if one CPU fails, the others keep going).
- **Clustered System:** Multiple independent computers connected over a network working together on one big task. Like multiple separate branch offices collaborating.

3. Operating System Structure

This is how the actual code of the OS is organized and built.

- **Simple Structure:** There are no clear boundaries. The applications can bypass the OS and talk directly to the hardware. (Example: MS-DOS). *Problem:* If one app crashes, the whole computer crashes.
- **Monolithic / Command-line:** Everything the OS does (file management, memory, CPU scheduling) is bundled into one massive block of code. It's very fast, but if there's a bug in one part, it can break everything.
- **Modular Structure (Loadable Kernel Modules):** Think of this like Lego blocks. The OS has a small core (the kernel) that handles the absolute basics. Everything else (like USB drivers or file systems) is added on as a separate "module" only when needed. It's highly flexible and stable.

4. Types of Operating System & Services (VVIMP)

Types of OS (How they handle tasks):

- **Batch OS:** You give the computer a batch of jobs. It does them one by one without any user interaction. (Like handing a stack of math tests to a grader).
- **Time-Sharing (Multitasking) OS:** The CPU switches between different tasks so quickly that it feels like everything is running at the same time. (What your PC or phone does right now).
- **Distributed OS:** Manages a group of separate computers to make them look like a single computer to the user.
- **Real-Time OS (RTOS):** Used when time is critical. A task *must* be done within an exact timeframe, or there is a failure. (Used in airbags, pacemakers, and rockets).

OS Services (What the OS does for you):

- **User Interface (UI):** Gives you a screen to click on (GUI) or a command line to type in.
- **Program Execution:** Loads your apps into memory and runs them.
- **I/O Operations:** Handles reading USBs, printing documents, or recording audio.
- **File-System Manipulation:** Lets you read, write, create, and delete files and folders.
- **Error Detection:** Constantly checks for hardware or software errors to prevent crashes.

5. System Calls & System Boot (VVIMP)

System Calls (The "Permission Slip"):

Programs are not allowed to directly touch hardware (for security). If a program wants to read a file, print a page, or access the camera, it must ask the OS to do it for them. **A System Call is the way a program formally requests a service from the Operating System.**

Types of System Calls:

1. **Process Control:** Create or terminate a process (e.g., closing an app).
2. **File Management:** Create, read, or delete a file.
3. **Device Management:** Request access to a printer or USB.
4. **Information Maintenance:** Get the current time or system data.
5. **Communications:** Send data over a network.

System Boot (Waking the computer up):

1. You press the Power button.
2. **BIOS/UEFI** runs: A tiny program on the motherboard checks all hardware to ensure it's working (POST - Power On Self Test).
3. **Bootloader:** The BIOS finds a small program on your hard drive called the bootloader.
4. **Load OS:** The bootloader grabs the Operating System (like Windows or Linux) from the hard drive and loads it into the RAM so it can start running.

6. Process States + Process Scheduling (VVIMP)

A **process** is simply a program in execution (like having Chrome open).

Process States (The Lifecycle):

As a process lives, it moves through different phases:

- **New:** The process is just being created.
- **Ready:** The process is loaded into memory and waiting for its turn to use the CPU.
- **Running:** The CPU is actively executing the process's instructions right now.
- **Waiting (or Blocked):** The process is paused because it's waiting for something to happen (like waiting for you to type a password or a file to download).
- **Terminated:** The process has finished its job or was closed.

Process Scheduling:

Since a computer has many processes but usually fewer CPU cores, the OS has to act like a traffic cop, deciding *who* gets to use the CPU and *when*.

- **Short-Term Scheduler (CPU Scheduler):** The fast decision-maker. It looks at the "Ready" queue and picks which process gets the CPU right this millisecond.
- **Long-Term Scheduler (Job Scheduler):** The slow decision-maker. It decides which programs are allowed into the "Ready" queue from the hard drive in the first place, ensuring the CPU doesn't get overwhelmed.